

DR ESLAM AHMED
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Expanding Brackets

Multiply every term by every term. Then collect. That is the whole topic — and you will use it in every algebra topic after this one.

THE TOPIC ON ONE MAP

Expanding Brackets

01 — One bracket

$3(2x - 5)$

$-2(x - 4)$

expand and simplify

02 — Two brackets

$(m - 8)(m + 5)$

the grid

03 — The special two

$(x + 3)^2$

$(a + b)(a - b)$

04 — Three brackets

$(x + 1)(x - 2)(x + 3)$

$ax^3 + bx^2 + cx + d$

05 — Where you will use it

show that ... = 0

single fraction

proof

Five ideas. After each idea you practise it straight away: my questions first, then selected practice, then real past papers.

01 One Bracket

The term outside multiplies **every** term inside — and the sign in front belongs to the term.

WHEN YOU SEE → YOU DO

“expand” / “multiply out”

→ outside term $\times$ every inside term

“expand and simplify”

→ expand, then collect like terms

“-(...)” or “-2(...)”

→ the minus changes every sign inside

$$-3 (2x - 5) = -6x + 15$$

1

Multiply

Outside term $\times$ each inside term. Take the sign with it.

2

Collect like terms

x terms together, numbers together. That is “simplify”.

WORKED MODEL • EXPAND AND SIMPLIFY

$$5(y - 2) - 2(y - 3)$$

$$\textcircled{1} \quad 5y - 10$$

first bracket

$$\textcircled{2} \quad -2y + 6$$

 $-2 \times -3 = +6$

$$\textcircled{3} \quad 5y - 2y - 10 + 6$$

line up like terms

$$\text{ANSWER} \quad 3y - 4$$

× **TRAP** A minus in front of a bracket changes **every** sign inside: $-(5c - 1) = -5c + 1$, not $-5c - 1$.

IDEA 01

Practise: One Bracket

My questions first, then selected practice.

Q1 **DR ESLAM** The sign goes with the term

CH 2 • TOPIC 3 2 marks

Expand (a) $-3(2x - 5)$ (b) $-x(4 - x)$

Q2 **DR ESLAM** Minus in front of a bracket

CH 2 • TOPIC 3 2 marks

Expand and simplify $4(x + 3) - 2(x - 5)$.

The -2 multiplies *both* terms: $-2 \times -5 = +10$.

Answer _____

Q3 **DR ESLAM** Letters outside

CH 2 · TOPIC 3 2 marks

Expand and simplify $2x(3x - 4) - x(x + 2)$.

Answer _____

SELECTED PRACTICE**Q4** **PRACTICE** Practice · Easy

CH 2 · TOPIC 3 1 mark

1 Expand $3(2y - 5)$

(1 mark)

Q5 **PRACTICE** Practice · Easy

CH 2 · TOPIC 3 1 mark

4 Expand $4(x + 2)$

(1 mark)

Q6**PRACTICE**

Practice · Easy

CH 2 · TOPIC 3 3 marks

7 (a) Expand $x(x + 2)$

(1 mark)

(b) Expand and simplify $3(y + 2) + 4(x - 1)$

(2 marks)

Q7**PRACTICE**

Practice · Easy

CH 2 · TOPIC 3 2 marks

11 Expand and simplify $3(x + 4) + 2(5x - 1)$

(2 marks)

Q8**PRACTICE**

Practice · Medium

CH 2 · TOPIC 3

2 marks

1 Expand and simplify $3(m+4) - 2(4m+1)$

(2 marks)

Q9**PRACTICE**

Practice · Medium

CH 2 · TOPIC 3

2 marks

16 Expand and simplify fully $4(2c+3) - (5c-1)$

(2 marks)

02 Two Brackets

Every term in the first bracket $\times$ every term in the second. Four products, then collect the middle two.

WHEN YOU SEE $\rightarrow$ YOU DO

$(\)(\)$

$\rightarrow$ grid: four products, collect the middle two

“circle the term which is part of the answer”

$\rightarrow$ only the middle terms are needed

$\times$	x	-4
$2x$	$2x^2$	$-8x$
$+3$	$+3x$	-12

$(2x + 3)(x - 4)$ — the gold cells are the like terms.

1

Fill the grid

Row $\times$ column in every cell.

2

Collect the gold cells

$$-8x + 3x = -5x$$

ANSWER $2x^2 - 5x - 12$

CALCULATOR CHECK Put $x = 2$ into the question and into your answer. $(7)(-2) = -14$ and $8 - 10 - 12 = -14$. Same number $\rightarrow$ correct. (On the calculator: type both with ALPHA X, press CALC, enter 2.)

$\times$ TRAP $(y + 4)(2 - y)$: the x^2 term is $-y^2$. Keep the order the question gives and watch the minus.

IDEA 02

Practise: Two Brackets

My questions first, then selected practice, then real past papers.

Q10 **DR ESLAM** Grid first

CH 2 · TOPIC 3 2 marks

Expand and simplify $(x + 5)(x - 2)$ using the grid.

×	x	+5
x		
-2		

Answer _____

Q11 **DR ESLAM** Numbers in front of x

CH 2 · TOPIC 3 2 marks

Expand and simplify $(3x - 2)(2x + 5)$.

Answer _____

SELECTED PRACTICE

Q12

PRACTICE

Practice · Easy

CH 2 · TOPIC 3 1 mark

15 $(2x - 4)(3x + 5)$ is expanded and simplified.

Circle the term which is part of the answer.

$2x$	$-2x$	$22x$	$-22x$
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(1 mark)

Q13

PRACTICE

Practice · Medium

CH 2 · TOPIC 3 2 marks

6 Expand and simplify $(x - 5)(x + 3)$

(2 marks)

Q14 **PRACTICE** Practice · Medium

CH 2 · TOPIC 3 2 marks

9 Expand and simplify $(m + 7)(m + 3)$

(2 marks)

Q15 **PRACTICE** Practice · Medium

CH 2 · TOPIC 3 2 marks

17 Expand and simplify $(m + 9)(m - 5)$

(2 marks)

PAST PAPERS

Q16**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q1

CH 2 · TOPIC 3 8 marks**You must write down all the stages in your working.**

1 $w = 5y^2 - y^3$

(a) Work out the value of w when $y = -2$ $w =$ (b) Factorise fully $8p^2 - 2p$

.....

(c) Expand $4t(3t - 2)$

.....

(d) Expand and simplify $(5x - 2)(x + 4)$

.....

(Total for Question 1 is 8 marks)

Q17**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q1

CH 2 • TOPIC 3**3 marks**

You must write down all the stages in your working.

1 (a) Simplify $e^8 \div e^2$

(b) Expand and simplify $(x - 3)(x + 1)$

(Total for Question 1 is 3 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q1

CH 2 • TOPIC 3**4 marks**

You must write down all the stages in your working.

1 (a) Expand and simplify $(y + 4)(2 - y)$

(b) Factorise fully $15b^5c - 35b^3c^9$

(Total for Question 1 is 4 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q1

CH 2 · TOPIC 3 4 marks**You must write down all the stages in your working.**

1 $P = m^2 - 4c$

(a) Work out the value of P when $m = -5$ and $c = 3$

$P = \dots\dots\dots$

(b) Expand and simplify $(x + 5)(x - 7)$

$\dots\dots\dots$

(Total for Question 1 is 4 marks)

03 The Special Two

Two patterns come back everywhere. Learn them so well you write the answer in one line.

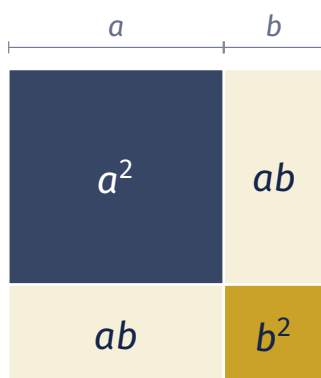
WHEN YOU SEE → YOU DO

"()²"

→ square · double · square

"(a + b)(a - b)", same numbers, opposite signs

→ the middle cancels: $a^2 - b^2$



$(a + b)^2$ is a square: a^2 , two ab strips, b^2 .

A

Perfect square

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

Square the first · double the product · square the last.

B

Difference of two squares

$$(a + b)(a - b) = a^2 - b^2$$

Same numbers, opposite signs → the middle terms cancel.

WORKED MODEL • PERFECT SQUARE IN ONE LINE

1 $(2x - 3)^2$

square · double · square

2 $(2x)^2 - 2(2x)(3) + 3^2$

ANSWER $4x^2 - 12x + 9$

× **TRAP** $(x + 3)^2 \neq x^2 + 9$. Where is the middle term? $(x + 3)^2 = x^2 + 6x + 9$.

IDEA 03

Practise: The Special Two

My questions first, then selected practice.

Q20 DR ESLAM Perfect-square drill

CH 2 • TOPIC 3 5 marks

Square the first • double the product • square the last.

(a) $(x + 4)^2$ (b) $(x - 6)^2$ (c) $(2x + 3)^2$ (d) $(3x - 1)^2$ (e) $\left(x + \frac{1}{2}\right)^2$

Q21 DR ESLAM Spot the mistake

CH 2 • TOPIC 3 2 marks

Sara writes $(x + 3)^2 = x^2 + 9$.

Explain her mistake and write the correct expansion.

Correct _____

Q22 **DR ESLAM** Difference of two squares

CH 2 • TOPIC 3 2 marks

Expand (a) $(x + 7)(x - 7)$ (b) $(2x - 5)(2x + 5)$

Same numbers, opposite signs: the middle terms cancel.

SELECTED PRACTICE**Q23** **PRACTICE** Practice • Easy

CH 2 • TOPIC 3 2 marks

16 Multiply out and simplify $(x - 8)^2$

(2 marks)

Q24 **PRACTICE** Practice • Medium

CH 2 • TOPIC 3 2 marks

13 Expand and simplify $(w - 5)^2$

(2 marks)

04 Three Brackets

The most common exam question on this topic: 13 of the 17 past-paper questions here have three brackets. Two first, then the third.

WHEN YOU SEE → YOU DO

three brackets

→ two first, then the third — single term last

“in the form $ax^3 + bx^2 + cx + d$ ”

→ expand fully, highest power first

“≡” / “work out the values of a, b, c ”

→ compare coefficients

1

Expand two brackets

Pick the two easiest. If one bracket is a single term like $5x$, leave it for last.

2

Multiply by the third

Every term $\times$ every term again — usually 6 products.

3

Collect, highest power first

Write it as $ax^3 + bx^2 + cx + d$.

WORKED MODEL • JUNE 2025 P2H STYLE

$$3x(2x + 5)(7x - 4)$$

$$1 \quad (2x + 5)(7x - 4) = 14x^2 - 8x + 35x - 20$$

two brackets

$$2 \quad = 14x^2 + 27x - 20$$

collect

$$3 \quad 3x(14x^2 + 27x - 20)$$

single term last

ANSWER $42x^3 + 81x^2 - 60x$

“SHOW THAT” / IDENTITIES Expand the left side fully, then **compare coefficients**: x^3 with x^3 , x^2 with x^2 , and so on.

IDEA 04

Practise: Three Brackets

My questions first, then selected practice, then real past papers.

Q25 DR ESLAM Single term last

CH 2 · TOPIC 3 3 marks

Expand and simplify $2x(x + 3)(x - 1)$.

Brackets first, then multiply everything by $2x$.

Answer _____

Q26 DR ESLAM Two first, then the third

CH 2 · TOPIC 3 3 marks

Expand and simplify $(x + 1)(x + 2)(x - 3)$.

Answer _____

SELECTED PRACTICE

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

- 1 Show that $(x + 1)(x + 2)(x + 3)$ can be written in the form $ax^3 + bx^2 + cx + d$ where a, b, c and d are positive integers.

(3 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 1 mark

- 11 Show that $x(x - 1)(x + 1) = x^3 - x$

(1 mark)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 4 marks

15 Expand and simplify $(3x^2 + 2)(2x + 5) - 6x(x^2 - 3)$

(4 marks)**Q30****PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

18 Expand and simplify $(x + 6)(3x - 2)(x + 6)$

(3 marks)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q14

CH 2 · TOPIC 3 3 marks

14 Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

(Total for Question 14 is 3 marks)

Q32 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 1HR · Q16

CH 2 · TOPIC 3 5 marks

16 (a) Expand and simplify $(x + 4)(x - 2)(x + 1)$ (b) Express $x^2 - 10x + 40$ in the form $(x + a)^2 + b$, where a and b are integers.

(Total for Question 16 is 5 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q12

CH 2 • TOPIC 3 6 marks

- 12 (a) Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.

Give your answer in its simplest form.

- (b) Expand and simplify $2x(x-5)(x-3)$

(Total for Question 12 is 6 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q13

CH 2 • TOPIC 3 3 marks

- 13 Expand and simplify $4x(3x+1)(2x-3)$
Show your working clearly.

(Total for Question 13 is 3 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q12

CH 2 • TOPIC 3 7 marks

12 (a) Simplify $(64p^9q^{12})^{\frac{2}{3}}$

.....

(b) Write as a single fraction $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$

Give your answer in its simplest form.

.....

(c) Expand and simplify $4x(x - 5)(2x + 3)$
Show your working clearly.

.....

(Total for Question 12 is 7 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q13

CH 2 • TOPIC 3 6 marks

13 (a) Expand and simplify $5x(x + 2)(3x - 4)$

.....
(3)

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

.....
(Total for Question 13 is 6 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q16

CH 2 • TOPIC 3 6 marks

16 (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$

.....
(3)

(b) Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$

.....
(Total for Question 16 is 6 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q16

CH 2 • TOPIC 3 3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

.....
(Total for Question 16 is 3 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q14

CH 2 • TOPIC 3 7 marks

- 14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(b) Solve $\frac{2x + 3}{5} + \frac{6x - 5}{4} = \frac{163}{100}$

Show clear algebraic working.

$x =$

(Total for Question 14 is 7 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q14

CH 2 • TOPIC 3 6 marks

14 (a) Expand and simplify $(3x + 1)(2 - x)(4 + x)$

.....
(3)

(b) Simplify fully $\left(\frac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$

.....
(Total for Question 14 is 6 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q15

CH 2 • TOPIC 3 3 marks

15 Write $3x(2x - 1)(5x + 4)$ in the form $ax^3 + bx^2 + cx$ where a , b and c are integers.

.....
(Total for Question 15 is 3 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q13

CH 2 • TOPIC 3 5 marks13 (a) Factorise $4x^2 - 25y^2$

(b) Show that $4x(x + 3)(2x - 5)$ can be written in the form $ax^3 + bx^2 + cx$ where a , b and c are integers to be found.

(Total for Question 13 is 5 marks)**Q43****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q12

CH 2 • TOPIC 3 3 marks12 Expand and simplify $3x(2x + 5)(7x - 4)$ **(Total for Question 12 is 3 marks)**

Q44**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q14

CH 2 • TOPIC 3 6 marks

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

$y =$

(Total for Question 14 is 6 marks)

05 Where You Will Use It

This topic is a tool. Every one of these later topics starts by expanding brackets.

WHEN YOU SEE → YOU DO

“show that ... = 0”

→ expand both sides, rearrange to = 0

“as a single fraction”

→ expand the numerator

“prove”

→ expand, then factorise to prove

Solving quadratics

ERCS: Expand first.

$$x(x - 5) = 14$$

Algebraic fractions

Adding: expand the top.

$$3(x + 1) - 2(x - 4)$$

Expanding brackets

Completing the square

Check by expanding back.

$$(x + 3)^2 + 2 = x^2 + 6x + 11$$

Proof & functions

$$(n + 3)^2 - (n - 3)^2 = 12n$$

fg(x), area “show that”

WORKED MODEL • SOLVE AFTER EXPANDING

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$1 \quad 4x^2 + 20x + 25$$

perfect square

$$2 \quad 4x^2 + 4x - 3$$

two brackets

$$3 \quad 20x + 25 = 4x - 3$$

the $4x^2$ cancel

ANSWER $x = -\frac{7}{4}$

IDEA 05

Practise: Where You Will Use It

My questions first, then selected practice.

Q45 **DR ESLAM** Quadratics: the E of ERCS

CH 2 • TOPIC 3 2 marks

Show that $(x + 2)^2 = 3x + 10$ can be written as $x^2 + x - 6 = 0$.

Q46 **DR ESLAM** Algebraic fractions: the numerator

CH 2 • TOPIC 3 3 marks

Write $\frac{3}{x-4} - \frac{2}{x+1}$ as a single fraction.

The top becomes $3(x + 1) - 2(x - 4)$ — expand and simplify it.

Answer _____

Q47 DR ESLAM Proof

CH 2 • TOPIC 3 3 marks

Show that $(n + 3)^2 - (n - 3)^2 = 12n$ for all values of n .

SELECTED PRACTICE**Q48** PRACTICE Practice • Hard

CH 2 • TOPIC 3 3 marks

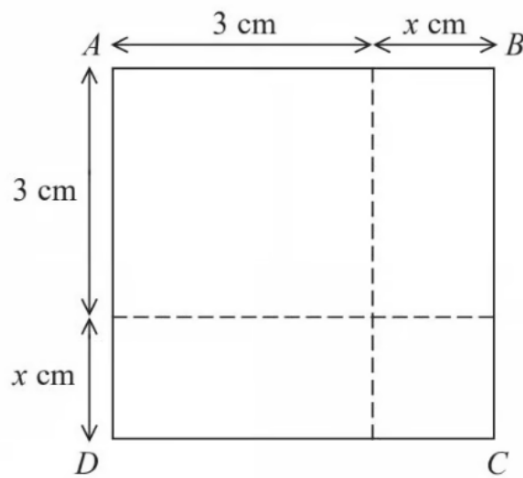
3 Simplify fully $(\sqrt{a} + \sqrt{4b})(\sqrt{a} - 2\sqrt{b})$

(3 marks)

Q49**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3

3 marks**5**

The area of square $ABCD$ is 10 cm^2 .

Show that $x^2 + 6x = 1$

(3 marks)

Q50**PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

- 7 Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer,

find the value of x and the value of y .

Show each stage of your working clearly.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(3 marks)**Q51****PRACTICE**

Practice · Hard

CH 2 · TOPIC 3 3 marks

- 8 Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x = \dots\dots\dots$

(3 marks)

Q52 **PRACTICE** Practice · Hard

CH 2 · TOPIC 3 3 marks

9 The functions f and g are defined as

$$f(x) = x^2 + 6$$

$$g(x) = x - 10$$

Solve the equation $fg(x) = f(x)$

Show clear algebraic working.

(3 marks)

Q53 **PRACTICE** Practice · Hard

CH 2 · TOPIC 3 3 marks

13 Here is an identity.

$$a(3x - 10) \equiv 21x + 2b$$

Work out the values of a and b .

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 One bracket** Outside $\times$ every term inside. The sign goes with the term.
- 02 Two brackets** Four products (grid), collect the middle two. Check with $x = 2$.
- 03 Special two** $(a \pm b)^2 = a^2 \pm 2ab + b^2$ $(a + b)(a - b) = a^2 - b^2$
- 04 Three brackets** Two first, then the third. Single term last. $ax^3 + bx^2 + cx + d$.
- 05 Use it** Quadratics (ERCS), fractions, completing the square, proof.

BOTTOM LINE Every term $\times$ every term $\rightarrow$ collect like terms $\rightarrow$ check with $x = 2$.

Answers

Expanding Brackets • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ONE BRACKET

- | | | |
|-----------|--|----------------------------------|
| Q1 | Dr Eslam • The sign goes with the term | (a) $-6x + 15$ (b) $-4x + x^2$ |
| Q2 | Dr Eslam • Minus in front of a bracket | $2x + 22$ |
| Q3 | Dr Eslam • Letters outside | $5x^2 - 10x$ |
| Q4 | Practice • Easy | $6y - 15$ |
| Q5 | Practice • Easy | $4x + 8$ |
| Q6 | Practice • Easy | (a) $x^2 + 2x$ (b) $3y + 4x + 2$ |
| Q7 | Practice • Easy | $13x + 10$ |
| Q8 | Practice • Medium | $10 - 5m$ |
| Q9 | Practice • Medium | $3c + 13$ |

IDEA 02 • TWO BRACKETS

- Q10** Dr Eslam • Grid first $x^2 + 3x - 10$
- Q11** Dr Eslam • Numbers in front of x $6x^2 + 11x - 10$
- Q12** Practice • Easy $-2x$ (full: $6x^2 - 2x - 20$)
- Q13** Practice • Medium $x^2 - 2x - 15$
- Q14** Practice • Medium $m^2 + 10m + 21$
- Q15** Practice • Medium $m^2 + 4m - 45$
- Q16** Edexcel 4MA1 • January 2021 • Paper 2HR • Q1 (a) 28 (b) $2p(4p-1)$ (c) $12t^2-8t$ (d) $5x^2+18x-8$
- Q17** Edexcel 4MA1 • November 2021 • Paper 1H • Q1 (a) e^6 (b) $x^2 - 2x - 3$
- Q18** Edexcel 4MA1 • January 2022 • Paper 2H • Q1 (a) $8 - 2y - y^2$ (b) $5b^3c(3b^2 - 7c^8)$
- Q19** Edexcel 4MA1 • June 2023 • Paper 2HR • Q1 (a) 13 (b) $x^2 - 2x - 35$

IDEA 03 • THE SPECIAL TWO

- Q20** Dr Eslam • Perfect-square drill (a) $x^2+8x+16$ (b) $x^2-12x+36$ (c) $4x^2+12x+9$
(d) $9x^2-6x+1$ (e) $x^2+x+\frac{1}{4}$
- Q21** Dr Eslam • Spot the mistake She missed the middle term $2 \times x \times 3$: x^2+6x+9
- Q22** Dr Eslam • Difference of two squares (a) $x^2 - 49$ (b) $4x^2 - 25$
- Q23** Practice • Easy $x^2 - 16x + 64$
- Q24** Practice • Medium $w^2 - 10w + 25$

IDEA 04 • THREE BRACKETS

- Q25** Dr Eslam • Single term last $2x^3 + 4x^2 - 6x$
- Q26** Dr Eslam • Two first, then the third $x^3 - 7x - 6$
- Q27** Practice • Hard $x^3 + 6x^2 + 11x + 6$
- Q28** Practice • Hard $x(x^2 - 1) = x^3 - x$
- Q29** Practice • Hard $15x^2 + 22x + 10$
- Q30** Practice • Hard $3x^3 + 34x^2 + 84x - 72$
- Q31** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q14 $20x^3 - 31x^2 - 81x - 18$
- Q32** Edexcel 4MA1 • January 2021 • Paper 1HR • Q16 (a) $x^3 + 3x^2 - 6x - 8$ (b) $(x - 5)^2 + 15$
- Q33** Edexcel 4MA1 • January 2021 • Paper 1H • Q12 (a) $\frac{x + 10}{(x - 2)(x + 1)}$ (b) $2x^3 - 16x^2 + 30x$
- Q34** Edexcel 4MA1 • May 2021 • Paper 1H • Q13 $24x^3 - 28x^2 - 12x$
- Q35** Edexcel 4MA1 • November 2021 • Paper 2H • Q12 (a) $16p^6q^8$ (b) $\frac{17}{30x}$ (c) $8x^3 - 28x^2 - 60x$
- Q36** Edexcel 4MA1 • January 2022 • Paper 1H • Q13 (a) $15x^3 + 10x^2 - 40x$ (b) $\frac{y^{15}}{8w^6}$
- Q37** Edexcel 4MA1 • June 2022 • Paper 1HR • Q16 (a) $9x^3 + 18x^2 - x - 2$ (b) $\frac{16y^4}{x^8}$
- Q38** Edexcel 4MA1 • November 2023 • Paper 2H • Q16 $2x^3 + x^2 - 43x - 60$
- Q39** Edexcel 4MA1 • June 2024 • Paper 2H • Q14 (a) $6y^3 + 57y^2 + 105y$ (b) $x = 1.2$
- Q40** Edexcel 4MA1 • May 2024 • Paper 1HR • Q14 (a) $-3x^3 - 7x^2 + 22x + 8$ (b) a^3b^2
- Q41** Edexcel 4MA1 • November 2024 $30x^3 + 9x^2 - 12x$

IDEA 05 • WHERE YOU WILL USE IT

Q45 Dr Eslam • Quadratics: the E of ERCS

$$x^2 + 4x + 4 = 3x + 10 \Rightarrow x^2 + x - 6 = 0$$

Q46 Dr Eslam • Algebraic fractions: the numerator

$$\frac{x + 11}{(x - 4)(x + 1)}$$

Q47 Dr Eslam • Proof

$$n^2 + 6n + 9 - (n^2 - 6n + 9) = 12n$$

Q48 Practice • Hard

$$a - 4b$$

Q49 Practice • Hard

$$(x + 3)^2 = 10 \Rightarrow x^2 + 6x = 1$$

Q50 Practice • Hard

$$40 + 3\sqrt{x} - x: x = 19, y = 3$$

Q51 Practice • Hard

$$x = -\frac{7}{4}$$

Q52 Practice • Hard

$$(x - 10)^2 + 6 = x^2 + 6 \Rightarrow x = 5$$

Q53 Practice • Hard

$$a = 7, b = -35$$